



Shri Vile Parle Kelavani Mandal's
DWARKADAS J. SANGHVI COLLEGE OF ENGINEERING
(Autonomous College Permanently Affiliated to the University of Mumbai)
NAAC Accredited with "A" Grade (CGPA : 3.18)



UNPLUGGED

A 24-HOUR HARDWARE HACKATHON



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ROUND 1

PROBLEM STATEMENT: DESIGN AND DEVELOP A HIGH-EFFICIENCY SCHEMATIC AND PCB LAYOUT USING KICAD WITHIN THE GIVEN TIME FRAME (4 HOURS) TO SERVE AS THE CENTRAL CONTROL SYSTEM FOR A LINE FOLLOWER CAR

PARTICIPANTS WILL BE PROVIDED WITH A PREDEFINED COMPONENT LIST, PCB DESIGN CONSTRAINTS, AND A REFERENCE IMAGE WITH EXACT DIMENSIONS OF THE LINE FOLLOWER CHASSIS. THE OBJECTIVE IS TO CREATE A COMPACT, OPTIMIZED PCB THAT FITS WITHIN THE SPECIFIED MECHANICAL BOUNDARIES WHILE ENSURING RELIABLE PERFORMANCE AND CLEAN SIGNAL ROUTING. THE PCB MUST BE DESIGNED TO ACCURATELY INTERFACE ALL PROVIDED COMPONENTS, ENSURING PROPER POWER DISTRIBUTION, SIGNAL INTEGRITY, AND EFFICIENT ROUTING UNDER GIVEN CONSTRAINTS. SPECIAL ATTENTION MUST BE GIVEN TO MAINTAINING DIMENSIONAL ACCURACY SO THAT THE PCB ALIGNS CLOSELY WITH THE PROVIDED CHASSIS REFERENCE (EITHER IDENTICAL OR FUNCTIONALLY EQUIVALENT IN SIZE AND MOUNTING COMPATIBILITY).

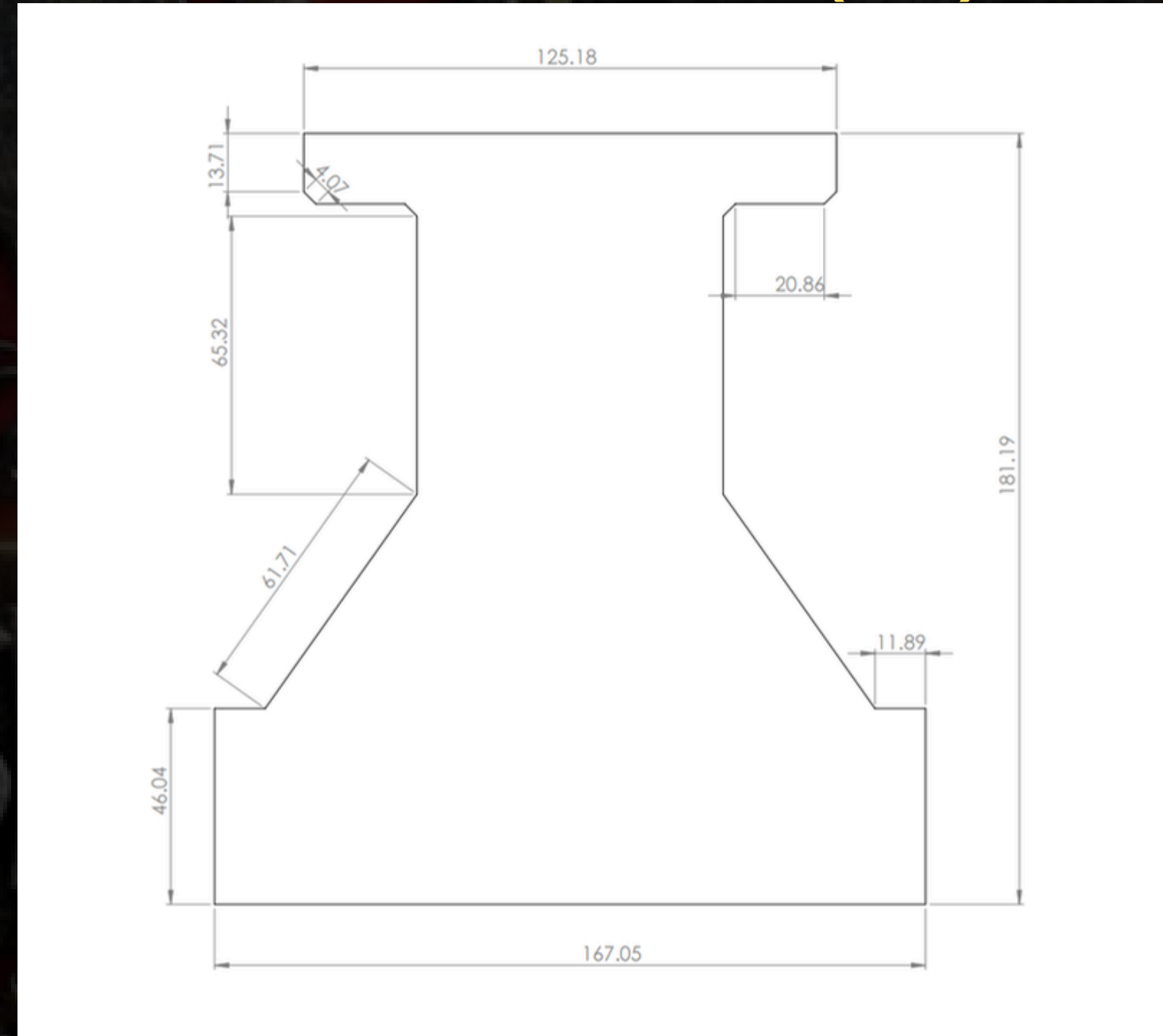
IN ADDITION TO THE PCB DESIGN, PARTICIPANTS ARE ENCOURAGED TO DEVELOP A COMPLETE CAD MODEL OF THE SYSTEM TO ENHANCE MECHANICAL INTEGRATION AND USABILITY. THE CAD DESIGN SHOULD DEMONSTRATE THOUGHTFUL ENGINEERING BY INCORPORATING STRUCTURAL, ACCESSIBILITY, AND THERMAL CONSIDERATIONS.

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COMPONENTS

Sr.No	Components
1	ESP32
2	ESP32 Camera Module
3	Motor Driver (TB6612)
4	12V DC Gear Motor
5	Neo 6M GPS Module
6	IR Sensor Module
7	Solar Panel
8	18650 Battery Holder
9	TP4056
10	Buck Module(LM296)
11	0.96' OLED display

DESIGN CONSTRAINTS(MM)



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BROWNIE POINTS

USING LTSPICE. THE OBJECTIVE IS TO DESIGN AND DEVELOP A **POLARITY-INDEPENDENT** INPUT INTERFACE SYSTEM CAPABLE OF ACCEPTING A DC VOLTAGE IN THE RANGE OF 12V TO 50V WITH UNKNOWN POLARITY, AND GENERATING A **REGULATED +12V OUTPUT ALONG WITH A POLARITY-INDICATING DIGITAL SIGNAL.** THE COMPLETE FUNCTIONALITY OF THE SYSTEM MUST BE DEMONSTRATED THROUGH SIMULATION RESULTS. THE SYSTEM SHOULD ACCEPT A WIDE-RANGE DC INPUT (12V TO 50V) IRRESPECTIVE OF POLARITY, **ENSURING SAFE AND RELIABLE OPERATION UNDER BOTH NORMAL AND REVERSE POLARITY CONDITIONS.** IT MUST PROVIDE A STABLE +12V DC OUTPUT SUITABLE FOR POWERING EXTERNAL CIRCUITS. THE DESIGN SHOULD ALSO GENERATE A LOGIC-LEVEL OUTPUT SIGNAL INDICATING INPUT POLARITY – PRODUCING 0V FOR NORMAL POLARITY AND 5V WHEN REVERSED. **ELECTRICAL ISOLATION BETWEEN THE INPUT AND OUTPUT STAGES MUST BE INCORPORATED TO ENHANCE SAFETY AND NOISE IMMUNITY. ALL REQUIREMENTS MUST BE VALIDATED THROUGH SIMULATION WAVEFORMS AND RESULTS.**

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JUDGING GUIDELINES / EXPECTED DELIVERABLES

- **BILL OF MATERIALS (BOM)**
- A COMPLETE **LIST OF ALL COMPONENTS** USED WITH SPECIFICATIONS
- **PCB DESIGN** SCREENSHOTS:
 - COMPLETE PCB LAYOUT
 - FRONT COPPER LAYER
 - BACK COPPER LAYER
 - 3D RENDERED VIEW OF THE PCB
 - SCHEMATIC DIAGRAM
- VERIFICATION CHECKS:
 - ELECTRICAL RULE CHECK (**ERC**)
 - DESIGN RULE CHECK (**DRC**)

- **CAD DELIVERABLES:**

- RENDERED IMAGES OF THE ENCLOSURE AND ASSEMBLY
- PCB INTEGRATION WITHIN THE CAD MODEL

- SIMULATION DELIVERABLES (**BROWNIE POINTS**):

- SCHEMATIC SCREENSHOTS
- SIMULATION SCREENSHOTS

PROJECT REPOSITORY:

A WELL-DOCUMENTED GITHUB REPOSITORY CONTAINING ALL DESIGN FILES, LIBRARIES, AND ASSETS USED IN THE PROJECT